ChatGPT

# Deep Research Task: Interpreting and Planning an Unspecified Request

## Executive Summary

When a user issues a **vague or empty research request**, the first task is to interpret what they *might* need. This report analyzes possible interpretations (e.g. literature review, market study, feasibility analysis, etc.), outlines how to approach each, and provides decision tools (flowchart, checklist) for refining the request. We detail for each *research intent* its scope, key questions, sources, methods, deliverables, effort, and cost, supported by expert guidelines and examples【9†L37-L43】【6†L152-L160】【14†L36-L44】【21†L23-L31】. The report also includes a decision flowchart (Mermaid) to help choose the right approach based on constraints (deadline, budget, audience) and a template checklist to clarify any vague task. In the absence of user constraints, a **default plan** is recommended – typically starting with a broad exploratory literature scan followed by targeted analysis. All major points are cited to authoritative sources.

## Possible Research Intents

An **unspecified "deep research"** request could mean many things. Common interpretations include:

- **Literature Review / Synthesis:** Surveying academic or technical literature on a topic.
- **Market / Competitive Analysis:** Researching industry trends, customer needs, competitors and market size.
- **Technical Feasibility Study:** Assessing whether a proposed project or technology can be implemented with given resources.
- **Data Collection & Analysis:** Gathering raw data (e.g. via surveys, web scraping, sensors) and analyzing it.
- **Policy Brief:** Summarizing evidence and making recommendations on a public policy issue.
- **Product Comparison / Benchmarking:** Comparing features, prices, or performance across alternatives.
- **Travel Itinerary Planning:** Researching destinations, logistics, and costs for a trip.
- **Grant Proposal Preparation:** Identifying funding sources and drafting a research or project proposal.

Each of these has very different **scope**, **audience**, and **outputs**. The sections below describe each intent in detail.

## Literature Review

- **Scope:** Gather and synthesize existing published research on a given topic. Focus is on *what is known* and *where gaps exist*【9†L37-L43】【9†L66-L74】.

- **Key Questions:** What has been studied? What are major findings or controversies? What gaps or open questions remain?
- **Sources/Databases:** Academic search engines and databases (e.g. Google Scholar, JSTOR, PubMed, IEEE Xplore, arXiv, discipline-specific indexes) for peer-reviewed papers and books【9†L37-L43】. Also consider theses, conference proceedings, and technical reports.
- **Search Strategy/Keywords:** Use broad and specific keyword combinations, Boolean operators, and citation chaining (looking at references of key papers). Refine by year, domain, and methodology.
- **Data to Collect:** Bibliographic metadata and abstracts of relevant publications; annotated notes on each source.
- **Methods of Analysis:** Thematic or chronological organization; summarizing, synthesizing, and critically evaluating the literature【9†L79-L88】【9†L119-L128】. Optionally, quantitative meta-analysis if data is available.
- **Deliverables/Formats:** A written report (often with sections like Introduction, Thematic summaries, Conclusions) or an annotated bibliography【9†L79-L88】【9†L108-L113】. May include tables or figures summarizing literature scope (e.g. number of studies by year, topic).
- **Time & Effort: Medium to High.** Thorough literature reviews can require weeks to months, depending on topic breadth and number of sources.
- **Estimated Cost (outsourcing): Low to Medium** (main cost is researcher time; usually no expensive resources needed). A skilled researcher or student might be hired.

For example, a literature review **"collects key sources on a topic and discusses them in conversation with each other"**【9†L37-L43】. It often identifies unanswered questions or theoretical frameworks【9†L66-L74】.

## Market Analysis

- **Scope:** Assess market size, demand, trends, customer segments, and competitor landscape for a product/service or industry.
- **Key Questions:** Is there demand for this product? How large is the potential market? Who are the customers and competitors? What are industry trends and economic indicators?【6†L152-L160】
- **Sources/Databases:** Secondary sources like industry reports (e.g. Gartner, Forrester), government data (census, labor statistics), trade association reports, and news articles【11†L219-L227】. Primary sources include surveys, interviews, or focus groups if specific insights are needed.
- **Search Strategy/Keywords:** Use terms like "market size for [industry]", "trends in [sector]", "consumer behavior [product]", or "[region] demographics" etc. Check both academic and business databases (e.g. IBISWorld, Statista, PubMed for health markets, etc).
- **Data to Collect:** Quantitative data (market statistics, sales figures, demographic data) and qualitative insights (customer interviews, expert opinions).
- **Methods of Analysis:** Combine **secondary research** for high-level context and **primary research** for specific questions【11†L183-L191】【11†L219-L227】. Techniques include SWOT analysis, PESTLE analysis, segmentation, and simple statistics or charts.
- **Deliverables/Formats:** Usually a structured report or presentation. It may include charts (market shares, growth curves), tables (competitor feature comparison), and a written narrative. Specific deliverables might be a "Market Research Report" or a slide deck with key findings.
- **Time & Effort: Medium to High.** Gathering and interpreting market data can be time-consuming (often weeks).

- **Estimated Cost (outsourcing): Medium to High.** Professional market research firms or consultants can charge thousands to tens of thousands of dollars for comprehensive studies.

Market research "is the systematic process of gathering, analyzing, and interpreting information about your target market, customers, and competitors"【11†L93-L100】. It typically answers strategic questions (e.g. product viability, competitive gaps)【11†L98-L100】. Primary research (surveys, interviews) is expensive and slow, while secondary research (reports, databases) is faster and cheaper【11†L188-L195】. Key questions from a U.S. Small Business Administration guide include demand, market size, economic indicators, location, saturation, and pricing【6†L152-L160】.

## Technical Feasibility Study

- **Scope:** Evaluate whether a specific project, product, or technology can be successfully developed and implemented with available resources and technology.
- **Key Questions:** Can this be built/implemented with current tools? What technical requirements exist? What resources (hardware, software, personnel) are needed? What is the timeline and cost? What risks or obstacles might there be?【14†L36-L44】
- **Sources/Databases:** Technical specifications, vendor documentation, academic/industry research on similar projects, case studies, and consultations with technical experts. For new tech, review standards and emerging tech roadmaps.
- **Search Strategy/Keywords:** Use terms like "feasibility study [technology]", "[technology] implementation case study", "requirements for [system]", and technical keywords specific to the domain (e.g. programming languages, platforms).
- **Data to Collect:** System requirements, component specs, benchmark data, cost estimates, and any existing proof-of-concept or prototype results.
- **Methods of Analysis:** System architecture analysis, prototype testing, cost estimation, and risk assessment. Often includes scenario modeling or small experiments.
- **Deliverables/Formats:** A feasibility report document, typically with sections as in [14]: Project Description, Technical Requirements, Technology Assessment, Implementation Feasibility, Cost Analysis, Risk Assessment. Figures might include architecture diagrams or timelines.
- **Time & Effort: High.** Thorough feasibility studies often involve detailed analysis or prototyping (weeks to months).
- **Estimated Cost (outsourcing): High.** May require hiring engineers or consultants; costs could range from thousands to tens of thousands depending on complexity.

According to industry guides, a technical feasibility study **"evaluates the practicality of a proposed project or system,"** determining if it can be developed with available tools【14†L36-L44】. It covers components like technical requirements, technology options, implementation plan, cost estimates, and risks【14†L71-L80】【14†L98-L107】.

## Data Collection & Analysis

- **Scope:** Acquire raw data (e.g. through experiments, surveys, web scraping, sensors, databases) and process it to answer questions or extract insights.
- **Key Questions:** What data is needed to answer the research question? Where can it be obtained? How to ensure quality? What analysis (statistical, ML, etc.) will be applied?

- **Sources/Databases:** Depends on domain: public data repositories (government open data, Kaggle), APIs, corporate databases, published datasets, or scraping websites. Also consider sensor or experimental data.
- **Search Strategy/Keywords:** Varies by data: e.g. “Open dataset [topic]”, “[domain] data repository”, “API for [service]”. For web data: identify relevant websites and use keywords to filter needed content.
- **Data to Collect:** Numerical tables, text corpora, images, etc., in raw form. May include cleaned/structured versions.
- **Methods of Analysis:** Data wrangling and cleaning, statistical analysis, data mining, visualization, machine learning modelling, depending on goals.
- **Deliverables/Formats:** Datasets in CSV/JSON, analysis scripts, and an analytical report or dashboard. Often includes charts and tables of summary statistics.
- **Time & Effort: Medium.** Time depends on data availability and cleaning needs; analysis complexity can vary widely.
- **Estimated Cost (outsourcing): Medium.** Data specialists or coders can be hired (rates vary).

For example, one might plan a data project by drafting a **data collection plan** and code for analysis (no single source to cite, but standard research methodology applies).

## Policy Brief

- **Scope:** Research a public policy issue and present findings and recommendations succinctly for decision-makers or stakeholders (often non-technical audience).
- **Key Questions:** What is the problem? What evidence exists? What policy options or interventions are supported by the data? Who are the stakeholders?
- **Sources/Databases:** Government reports, policy think-tanks, white papers, relevant academic studies. Official statistics and previous policy analyses are primary sources.
- **Search Strategy/Keywords:** “[Policy issue] policy brief”, “[issue] government report”, “[issue] statistics”. Also try non-academic databases like World Bank, WHO, etc., for policy facts.
- **Data to Collect:** Key facts and figures (demographics, trends), results of relevant studies, expert opinions or interviews if needed.
- **Methods of Analysis:** Synthesize evidence to frame the problem, analyze impacts, and compare policy options. Use concise and clear language (avoid jargon)【21†L67-L75】.
- **Deliverables/Formats:** A short (often 1–2 page) document or PDF. Components include: executive summary, context/background, analysis of evidence, and recommendations【21†L98-L107】. Bullets and headings are common.
- **Time & Effort: Medium.** Requires careful writing to be concise; research itself may be less extensive than a full report.
- **Estimated Cost (outsourcing): Medium.** Policy analysts or consultants charge for brief preparation.

A policy brief is **“a concise summary of information [to] help readers understand… an issue and likely make decisions”**【21†L25-L31】. It should be laser-focused on one issue, using plain language and strong evidence【21†L23-L31】【21†L69-L77】. Typical sections include an executive summary, background/analysis, and conclusions or recommendations【21†L98-L107】.

## Product Comparison

- **Scope:** Compare multiple products or services (e.g. software, gadgets) to evaluate which best meets certain criteria.
- **Key Questions:** How do options differ in features, price, performance? What are pros and cons of each? Which provides the best value or fit?
- **Sources/Databases:** Manufacturer/product websites (for specs), user reviews (e.g. Amazon, CNET), independent reviews or benchmarks (tech blogs, industry publications), and potentially company financial filings for context.
- **Search Strategy/Keywords:** "Compare [Product A] vs [Product B] features", "[Product] review", "[product category] features". Also use price/feature filters on e-commerce or comparison sites.
- **Data to Collect:** Feature lists, prices, performance metrics (e.g. battery life, speed), user ratings, test results.
- **Methods of Analysis:** Tabulate features across products, highlight key differentiators, calculate summary ratings, and possibly expert judgement.
- **Deliverables/Formats:** Comparison table or spreadsheet, accompanied by narrative summary. Possible output is a slide deck or report with recommendations. Visual charts (e.g. bar chart of specs) may be included.
- **Time & Effort: Low to Medium.** Depends on number of products and depth of analysis; can be done in days for a few products.
- **Estimated Cost (outsourcing): Low to Medium.** Might hire a freelancer for a simple comparison (hundreds of dollars), or more if expert testing is needed.

Product comparisons often result in matrices or ranking reports. No single authoritative citation, but this approach is standard in consumer research. As an example, one travel blog notes organizing product lists in "neat compartments" to compare easily【27†L39-L48】.

## Travel Planning

- **Scope:** Research and organize the details of a trip (destinations, itinerary, transportation, lodging, activities, costs).
- **Key Questions:** Where to go and when? What attractions or experiences are priorities? What is the budget? What logistics (flights, visas, vaccines) are needed?
- **Sources/Databases:** Travel guides (Lonely Planet, Fodor's), official tourism websites, flight/hotel aggregator sites (Google Flights, Expedia), travel blogs and forums (e.g. TripAdvisor, Reddit).
- **Search Strategy/Keywords:** "[City] things to do", "[Country] travel visa requirements", "best hotels in [city]", "itinerary [destination] [days]". Use Google Maps for logistics and Google Flights for prices.
- **Data to Collect:** Visa requirements, accommodation options and rates, flight schedules, attraction hours and ticket prices, local transportation info, weather/climate data.
- **Methods of Analysis:** Build an itinerary and budget. Tools like spreadsheets or itinerary apps can help. Prioritize based on interests and time.
- **Deliverables/Formats:** A proposed itinerary document or slideshow with dates, locations, activities, and budget. Could include a map, schedule, and a packing or planning checklist.
- **Time & Effort: Medium.** Depending on trip complexity; a week-long trip with many destinations can take several days of planning.
- **Estimated Cost (outsourcing): Low to Medium.** Travel agents or planners may charge a fee (hundreds of dollars) or get commission on bookings.

A travel planning guide suggests using checklists to ensure you research **flights, accommodations, transport, and activities**【27†L113-L120】. Organization tools (maps, saved documents) are recommended to compile the information【27†L54-L63】【27†L122-L132】.

## Grant Proposal

- **Scope:** Identify funding opportunities and develop a project proposal to secure grants for research or initiatives.
- **Key Questions:** What is the specific research or project idea? Which agencies or foundations fund this area? How to frame objectives, methods, and impact to funders? What budget and timeline are needed?
- **Sources/Databases:** Grants databases (e.g. grants.gov, GrantForward, Foundation Directory), funding agency websites, and previous funded project examples. Institutional research offices often have resources.
- **Search Strategy/Keywords:** "Research grant [field]", "funding for [topic]", "small business innovation grant", "[agency name] grant programs".
- **Data to Collect:** Grant guidelines, eligibility criteria, submission deadlines, and examples of successful proposals.
- **Methods of Analysis:** Align project goals with funder priorities, draft proposal sections (aims, significance, methods), estimate budget, and ensure compliance with application rules.
- **Deliverables/Formats:** A complete proposal package: narrative (with sections like objectives, background/lit review, methods, expected outcomes), budget and justification, CVs of key personnel, and any required forms. Often delivered as a PDF or online submission.
- **Time & Effort: High.** Quality proposals require extensive preparation (often months).
- **Estimated Cost (outsourcing): High.** Professional grant writers may charge $50–$150 per hour or fixed fees (possibly thousands), especially for large grants.

Grant proposals are formal documents aimed at obtaining funding【31†L19-L27】. They typically include an introduction, literature review, methods, goals/outcomes, budget, and timeline【35†L200-L208】【35†L259-L262】. Reviewers expect a clear research plan and alignment with funding priorities.

## Comparison of Research Paths

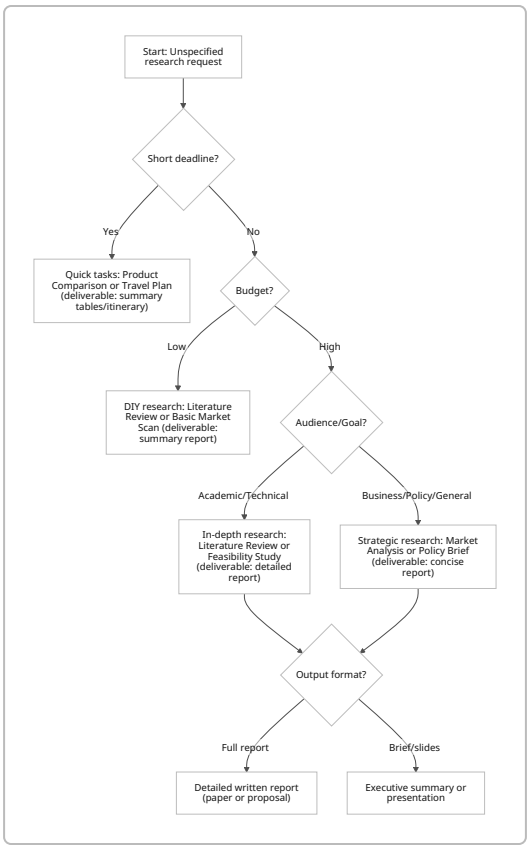
| Intent / Task | Scope | Typical Deliverables | Time/ Effort | Approx. Cost (outsourced) |
|---|---|---|---|---|
| **Literature Review** | Academic survey of prior work | Written review report, annotated bibliography【9†L37-L43】 | Medium–High | Low–Medium |
| **Market Analysis** | Industry/market trends & size | Report/slides, charts, SWOT/PESTLE【6†L152-L160】【11†L93-L100】 | Medium–High | Medium–High |
| **Technical Feasibility** | Technical/project viability | Feasibility report (sections: reqs, cost, risks)【14†L71-L80】 | High | High |

| Intent / Task | Scope | Typical Deliverables | Time/ Effort | Approx. Cost (outsourced) |
|---|---|---|---|---|
| **Data Analysis** | Collect & analyze raw data | Data set, analysis report, visualizations | Medium | Medium |
| **Policy Brief** | Policy problem and recommendations | 1–2 page brief, executive summary【21†L98-L107】 | Medium | Medium |
| **Product Comparison** | Feature/price/ performance comparison | Comparison table, summary report | Low–Medium | Low–Medium |
| **Travel Planning** | Trip itinerary and logistics | Itinerary document, budget overview | Medium | Low–Medium |
| **Grant Proposal** | Funding application for a project | Proposal narrative, budget, timeline【35†L200-L208】 | High | High |

*Table: Comparison of possible research tasks by scope, deliverables, effort, and cost. (Deliverables examples are indicative.)*

## Decision Flowchart

The flowchart below helps select an appropriate research approach given constraints like deadline, budget, audience, and desired output. For example, with a *short deadline* one might choose a quick survey or comparison, whereas a *high-budget, deep audience* scenario might allow a thorough literature review or feasibility study.




## Checklist for Defining Your Research Brief

Use this checklist to **turn a vague request into a clear research brief**. Fill in details as applicable:

- **Topic/Question:** What specific question or topic should the research address?
- **Purpose/Goal:** Why is this research needed? (Inform strategy, solve a problem, prepare a report, etc.)
- **Audience:** Who will use the results? (Technical experts, executives, public officials, general public?)
- **Scope & Depth:** Broad survey vs. in-depth analysis? Geographic or domain limits?
- **Output Format:** What deliverable is expected? (Detailed report, slide deck, executive summary, dataset, etc.)
- **Deadline:** When is it needed? Is there urgency?
- **Budget/Resources:** Any constraints on time, money, or tools? Will external help be used?
- **Key Questions:** What specific sub-questions should be answered?
- **Relevant Sources:** Are there known sources or datasets to use or avoid?
- **Constraints:** Any limitations (confidentiality, accessibility of data, etc.)?

Filling in these items will guide the search strategy, depth of analysis, and format of deliverables.

## Recommended Next Steps and Default Plan

**Next Steps:** If the user cannot or does not specify details, a good default approach is to start broad and then focus. For example: 1. **Clarify scope via checklist:** Attempt to answer the above checklist yourself or

by asking whoever set the task (if possible). 2. **Initial Literature/Background Scan:** Begin with a quick survey of existing information. For instance, do a preliminary search on Google Scholar and Google to see what comes up on the topic and see which of the intents (literature review, market data, etc.) is most relevant. 3. **Identify Gaps or Needs:** Based on the scan, determine what's missing. This may reveal the user's true goal (e.g. if a market report is scarce, the user might need market analysis). 4. **Choose a Research Path:** Use the flowchart logic to pick an approach. If unconstrained, a *literature review* or *market overview* is often a safe default – it builds foundational knowledge for most topics. 5. **Gather Core Sources:** Collect 5–10 key references or data sources to anchor the research (academic papers, industry reports, statistical databases). 6. **Plan the Deliverable:** Decide on the structure (e.g. report outline) and communicate it.

**Default Plan (assuming no constraints):**

- **Step 1:** Conduct a broad literature search to learn what content exists on the topic.
- **Step 2:** Summarize high-level findings (scope of existing knowledge or market).
- **Step 3:** Identify 2–3 specific angles (e.g. technical challenge, market segment, policy issue) and gather more detailed data for those.
- **Step 4:** Draft a structured report or presentation with an executive summary, background, analysis, and recommendations.

This default approach yields a comprehensive foundation. If later the user clarifies their intent (e.g. they only needed a quick market snapshot), parts of the work can be trimmed. Starting broadly ensures no obvious avenue is missed, and outputs (bibliography, data summaries) can be reused for the final product.

**Assumptions:** We assume here that time and budget are not extremely limited. If constraints are revealed (short deadline, low budget), adapt by narrowing scope (e.g. produce a summary of existing reports rather than new data collection).

**Sources:** The strategies above are drawn from standard research guides and industry resources【9†L37-L43】【6†L152-L160】【14†L36-L44】【21†L23-L31】【35†L200-L208】. These emphasize systematic planning (define goals/questions), using appropriate sources (academic vs. business vs. official data), and tailoring the deliverable to the audience.

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